



14.2.6 Wrap-Up: Reflection

1. Complete the learning pyramid.

One question I would like to have answered...	
One mistake I learned from today...	One thing I am not sure about...YET!
One thing I am confident I know is...	



Unit 14, Lesson 3: Discovering Inverse Operations



Benchmark

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.



Learning Targets

- ☐ I can use manipulatives to discover the role of inverse operations when solving one-step addition and subtraction equations.
- ☐ I can solve addition and subtraction one-step equations in one variable using manipulatives, drawings, and number lines to show the inverse operations.



1. Sketch a picture to represent this situation.

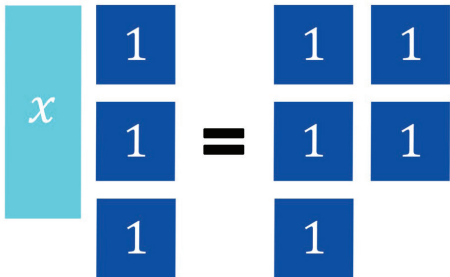
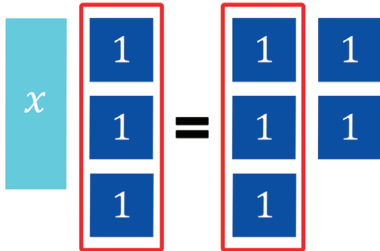
2. How much farther does Dale have to go to reach the roof? Include units.



In the model shown, **1** represents 1 and **x** represents x .

1. The steps to determine the value of x using algebra tiles are shown.

- Work with your partner to complete the last step according to the description.

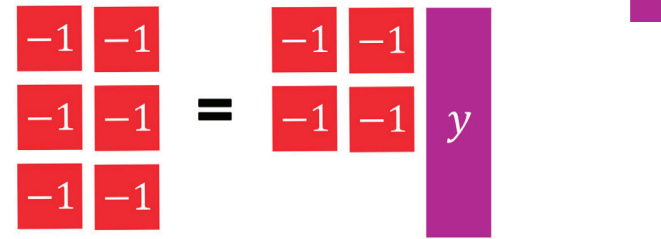
Algebra Tiles	Description
	Use algebra tiles to represent the equation $x + 3 = 5$.
	Determine any tiles that are represented on both sides of the equation.
	Remove the equal parts from both sides of the equation.

- b. The expressions $x + 3$ and 5 are equivalent according to the given equation. Discuss with your partner whether the tiles you drew on each side of the equation mat in the last step represent an equivalent relationship. Then, write a brief summary of your discussion.

- c. How many blue tiles are equivalent to one teal tile?

- d. What is the value of x in the equation $x + 3 = 5$?

2. In the model shown, -1 represents -1 and y represents y .



- a. Work with your partner to determine how many red tiles are equivalent to one purple tile.

- b. What is the value of y in the equation $-6 = -4 + y$?



14.3.3 Guided Instruction: Solving Equations Using Inverse Operations

An equation relates two equivalent expressions. When solving an equation, the mathematical sentence must remain balanced so that the two expressions remain equivalent.

1. Discuss with your partner what it means for an equation to remain balanced. Then, summarize your discussion.

When finding the solution to an equation, inverse operations are used to balance an equation.

An **inverse operation** is an operation that reverses the effect of another operation.

2. Using the definition of inverse operations, complete the table to indicate which operation would reverse the given operation.

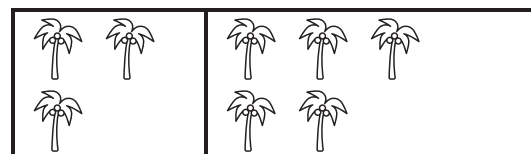
Operation	Inverse Operation
Subtraction (−)	
Addition (+)	



14.3.4 Guided Instruction: Solving Equations With Inverse Operations

Bar models and balance scales are useful tools for representing equations.

1. In the bar model shown, each palm tree represents 1 unit.

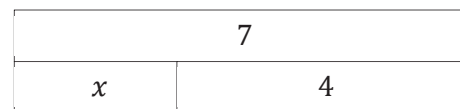


The bar model can be used to represent either an addition or subtraction equation. Fill in the missing values in each equation to match the bar model.

$$3 + \underline{\quad} = 8$$

$$8 - 3 = \underline{\quad}$$

2. The bar model shown can be used to represent either an addition or subtraction equation.



- a. Circle the addition equation that matches the diagram.

$$x + 7 = 4$$

$$7 = x + 4$$

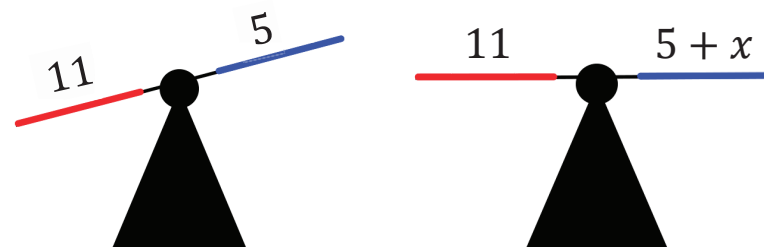
$$x = 7 + 4$$

b. Write an equivalent equation using the inverse operation.

c. Justify to your partner that the equation you wrote is an equivalent equation.

d. What is the value of the variable, x ?

3. Two balance scales are shown. The scale on the left is unbalanced. To balance the scale, an unknown value, x , needs to be added to the right side of the scale.



a. The equation represented by the balance scale on the right is shown. Use inverse operations to find the value of x by completing the steps shown.

$$x + 5 = 11$$

$$x + 5 - \square = 11 - \square$$

$$x = \square$$

b. Rewrite the original equation, substituting the solution for x .

c. Explain how to use the equation in Part B to confirm that your solution is correct.

Inverse operations can be used to find the value of the variable in an algebraic equation by creating equivalent equations.



Addition Property of Equality

If $a = b$, then $a + c = b + c$

Subtraction Property of Equality

If $a = b$, then $a - c = b - c$

4. The equation $x + 23 = 78$ can be solved by using inverse operations.

- a. Fill in values in each equivalent equation to solve for x .

$$x + 23 = 78$$

$$x + 23 - \underline{\hspace{2cm}} = 78 - \underline{\hspace{2cm}}$$

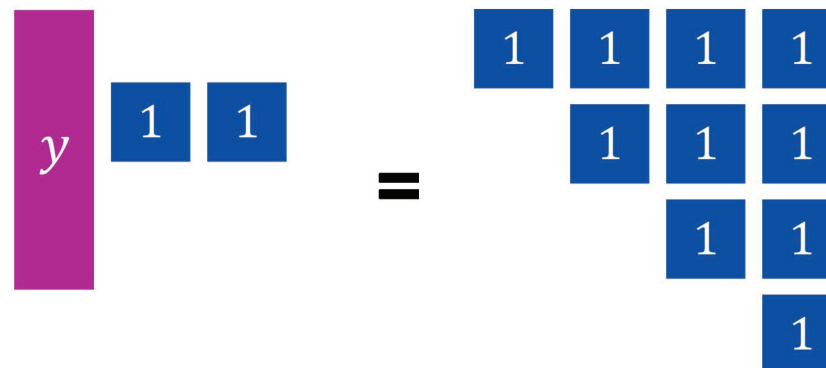
$$x = \underline{\hspace{2cm}}$$

- b. What is the inverse operation used to solve the equation $x + 23 = 78$?
- c. With your partner, discuss how to confirm whether or not the solution is accurate.



14.3.5 Your Turn!

1. The algebra tiles below model an equation.



- a. How many blue tiles are equivalent to one purple tile?
- b. Write a one-variable equation to represent the tiles on the mat.
- c. What inverse operation can be used to solve the equation?
- d. What value of y makes the equation true?

2. The equation $-31 = s - 6$ can be solved using inverse operations.

a. Fill in the missing information to solve for s .

$$-31 = s - 6$$

$$-31 + \underline{\hspace{2cm}} = s - 6 + \underline{\hspace{2cm}}$$

$$\underline{\hspace{2cm}} = s$$

b. Which inverse operation was used to solve the equation?

c. Check your solution.

3. Two bar model diagrams are shown.

Bar Model Diagram 1	Bar Model Diagram 2								
<table><tr><td colspan="2">m</td></tr><tr><td>-12</td><td>-4</td></tr></table>	m		-12	-4	<table><tr><td colspan="2">-12</td></tr><tr><td>4</td><td>m</td></tr></table>	-12		4	m
m									
-12	-4								
-12									
4	m								

a. Write an algebraic equation for each diagram box diagram.

Algebraic Equation for Diagram 1	Algebraic Equation for Diagram 2

b. Find the value of variable, m , for each algebraic equation.

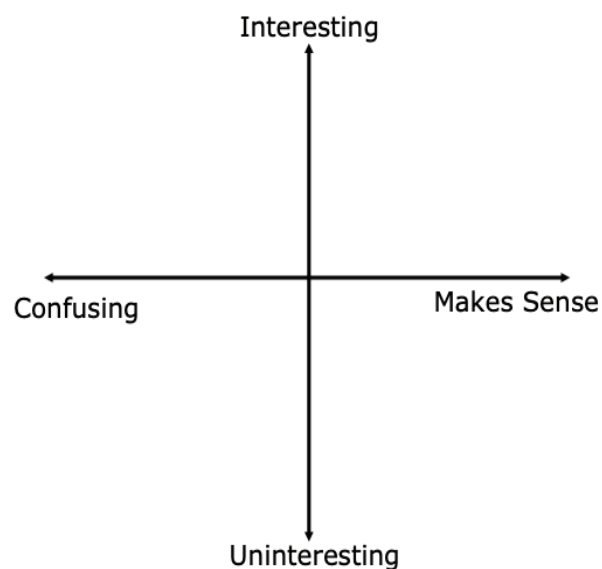
Diagram 1	Diagram 2

c. Discuss with your partner what you notice about the two equations. Summarize your discussion.



14.3.6 Wrap-Up: Reflection

1. Plot a point to indicate how you feel about today's learning target: "I can use diagrams and inverse operations when solving one-step addition and subtraction equations."



Unit 14, Lesson 4: Solving One-Step Equations in One Variable Using Addition and Subtraction



Benchmark

MA.6.AR.2.2 Write and solve one-step equations in one variable within a mathematical or real-world context using addition and subtraction, where all terms and solutions are integers.



Learning Target

- ☐ I can solve addition and subtraction one-step equations.